

TASK-3: To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables

```
root@LAPTOP-RJFDBEST:~# cd
root@LAPTOP-RJFDBEST:~# mkdir -p project/src project/include project/bin project/docs
root@LAPTOP-RJFDBEST:~# mv pipeline_jobs.c project/src/
root@LAPTOP-RJFDBEST:~# mv pipeline_jobs project/bin/ 2>/dev/null
root@LAPTOP-RJFDBEST:~# cd project
root@LAPTOP-RJFDBEST:~/project# find . -maxdepth 2 -type f
./src/pipeline_jobs.c
root@LAPTOP-RJFDBEST:~/project# nano Makefile
root@LAPTOP-RJFDBEST:~/project# make
Make: Nothing to be done for 'all'.
root@LAPTOP-RJFDBEST:~/project# ./bin/pipeline_jobs
=====
MULTI-JOB PIPELINE PERFORMANCE TEST
=====
Job 1 started: PID = 2088
Job 1 started: PID = 2091
Job 1 started: PID = 2089
All 1 jobs launched.
Job 1 started: PID = 2093
Job 4 started: PID = 2092
Job 2 completed: Result = 2088000000
Job 3 completed: Result = 2090000000
PID 2089: Job completed successfully.
PID 2090: Job failed with exit status 1.
PID 2091: Job completed successfully.
PID 2092: Job completed successfully.
PID 3 completed: Result = 2075000000
PID 2093: Job completed successfully.
=====
PERFORMANCE REPORT
=====
Total Jobs: 5
Successful Jobs: 4
Failed Jobs: 1
Total Time: 0.213 seconds
User CPU Time: 0.923 seconds
System CPU Time: 0.084 seconds
Maximum RSS: 1076 KB
Overall Status: FAILURE DETECTED
=====
root@LAPTOP-RJFDBEST:~/project# nano README.md
root@LAPTOP-RJFDBEST:~/project# make README.md
root@LAPTOP-RJFDBEST:~/project# find . -maxdepth 2 -type f
./src/pipeline_jobs.c
./bin/pipeline_jobs
./Makefile
./README.md
root@LAPTOP-RJFDBEST:~/project# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /root/project/.git/
root@LAPTOP-RJFDBEST:~/project# git status
On branch master
No commits yet
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   README.md
    new file:   bin/pipeline_jobs
    new file:   src/pipeline_jobs.c
root@LAPTOP-RJFDBEST:~/project# git commit -m "Organize project and add build system"
[master 35283e2] Organize project and add build system
Committer: Your Name <root@LAPTOP-RJFDBEST.localdomain>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:
    git config --global user.name "Your Name"
    git config --global user.email you@example.com
After doing this, you may fix the identity used for this commit with:
    git commit --amend --reset-author
4 files changed, 215 insertions(+)
create mode 100644 Makefile
create mode 100644 bin/pipeline_jobs
create mode 100644 src/pipeline_jobs.c
root@LAPTOP-RJFDBEST:~/project# git tag -a v1.0 -m "Initial stable release"
root@LAPTOP-RJFDBEST:~/project# git tag
v1.0
root@LAPTOP-RJFDBEST:~/project# git status
On branch master
nothing to commit, working tree clean
root@LAPTOP-RJFDBEST:~/project# ls -lh
-rw-r--r-- 1 root root 298 Sep 27 18:15 Makefile
-rw-r--r-- 1 root root 297 Sep 27 18:14 README.md
-rwxr-xr-x 2 root root 4.0k Sep 27 18:18 bin
-rwxr-xr-x 2 root root 4.0k Sep 27 18:12 docs
-rwxr-xr-x 2 root root 4.0k Sep 27 18:12 include
-rwxr-xr-x 2 root root 4.0k Sep 27 18:13 src
root@LAPTOP-RJFDBEST:~/project#
```

TASK-4:

TO Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review

```
root@LAPTOP-RJFDBEST:~/project# nano signal_demo.c
root@LAPTOP-RJFDBEST:~/project# gcc -Wall -Wextra signal_demo.c -o signal_demo
root@LAPTOP-RJFDBEST:~/project# ./signal_demo
Signal demonstration started.
PID = 606
Press Ctrl+C to send SIGINT.
Program is running...
^C
SIGINT received. Program handled the signal.
Program is running..
Program is running..
Program is running...
^Z
[1]+  Stopped                  ./signal_demo
root@LAPTOP-RJFDBEST:~/project# jobs -l
[1]+  606 Stopped              ./signal_demo
root@LAPTOP-RJFDBEST:~/project# kill -9 <PID>
-bash: syntax error near unexpected token 'newline'
root@LAPTOP-RJFDBEST:~/project# jobs
[1]+  Stopped                  ./signal_demo
root@LAPTOP-RJFDBEST:~/project# git add .
root@LAPTOP-RJFDBEST:~/project# git commit -m "Complete final project demonstration"
[master 35283e2] Complete final project demonstration
Committer: Your Name <root@LAPTOP-RJFDBEST.localdomain>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:
    git config --global user.name "Your Name"
    git config --global user.email you@example.com
After doing this, you may fix the identity used for this commit with:
    git commit --amend --reset-author
3 files changed, 84 insertions(+)
create mode 100644 demo_output.txt
Program is running...
Program is running...
^Z
[1]+  Stopped                  ./signal_demo
root@LAPTOP-RJFDBEST:~/project# jobs -l
[1]+  606 Stopped              ./signal_demo
root@LAPTOP-RJFDBEST:~/project# kill -9 <PID>
-bash: syntax error near unexpected token 'newline'
root@LAPTOP-RJFDBEST:~/project# jobs
[1]+  Stopped                  ./signal_demo
root@LAPTOP-RJFDBEST:~/project# git add .
root@LAPTOP-RJFDBEST:~/project# git commit -m "Complete final project demonstration"
[master 35283e2] Complete final project demonstration
Committer: Your Name <root@LAPTOP-RJFDBEST.localdomain>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:
    git config --global user.name "Your Name"
    git config --global user.email you@example.com
After doing this, you may fix the identity used for this commit with:
    git commit --amend --reset-author
3 files changed, 84 insertions(+)
create mode 100644 demo_output.txt
```